Careers in Information Technology: Computer & Information Research and Software Developer

Gerson Flores Fuentes

Northern Virginia Community College

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Prof. Nasser Maksoud

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Introduction

For this research assignment, I have decided on two information technology careers. The first one is Computer and Information Research Scientist, and the second is Software Developer. I pulled a list of comparisons between them on education, projected demand, daily tasks, and working conditions for each. I narrowed my list to computer and information research scientist and software developer from a long list of information technology occupations on the U.S. Bureau of Labor Statistics website, including O*NET for more insight on working conditions. Both require expertise and a variety of skills to perform at a professional level. I have always wondered how apps work on modern hardware, and the development of artificial intelligence made me curious about the technology and the various uses behind it.

Computer and Information Research Scientist

Computer and information research scientists design practical uses for computing technology, and they solve very complex problems in business, science, and medicine. They also develop theories and models, run experiments using data science and machine learning, and publish their research. One typical entry-level education is a master's degree in computer science, and some employers prefer a PhD; however, the federal government hires some candidates with a bachelor's degree; no license, exam, or certification is required. For this occupation, the starting median wage was $140,300 per year or $67.45 per hour. In May 2025, pay ranged from $85,460 at state universities to $211,270 at software publishers, and because most jobs require a graduate degree, education clearly affects earnings. Employment is projected to grow 22% from 2025 to 2035, driven by AI and cybersecurity research; however, the field is very small, with about 2900 openings per year, and 31% of the jobs are in federal government (BLS, 2026). Research

scientists can also expand their expertise to other computer and information systems management roles, which can carry responsibilities like directing staff and managing budgets (ONET, 2026). Continuing education is very necessary in this field because it is a continuously growing industry. A day in the life of a scientist might involve checking emails, meeting with engineers about why a model is underperforming, and spending the entire afternoon testing an algorithm and drafting a conference paper. On a surface level, I find the work to be very rewarding, with a great salary and career growth opportunities. But what is less exciting to me are the years of graduate school, the small number of openings, and hours spent working in front of a screen.

Software Developer

Software developers create applications for multiple uses and build systems tailored to operate devices and networks (BLS, 2026). They analyze users' needs, plan how a program's components will work together, and test and maintain software. Developers typically need a bachelor's degree in computer and information technology or a related field, and internships can provide valuable experience. No license or certification is required. Approximately 85% of respondents reported that a bachelor's degree is required. In May 2025, the median wage for developers was $135,980 per year, or $65.38 per hour, while software publishers paid a median of $164,550 (BLS, 2026). Some employers may prefer candidates with a master's degree, so education can influence hiring. Developers hold about 1.7 million jobs, and employment is projected to grow 10% by 2035. The main drawbacks are prolonged sitting, reported by 95% of workers, and a competitive job market. In addition, developers can be promoted to project management positions or computer and information systems manager roles; this could include being a supervising programmer and reporting project progress (ONET). Project management

training may help, and continuing education is required to keep up with new tools and languages. A normal day might start with a team meeting, followed by writing and testing features, reviewing a teammate's code, and fixing reported defects; 71% of developers say teamwork is extremely important (ONET, 2026). Although I find it very surprising that this job requires only a bachelor's degree, it offers many opportunities across industries. The downsides to this occupation are sitting most of the day, the highly competitive environment, and the constant demand to learn new technology.

Conclusion

Both careers are very well paid, with medians of 140,300 and $135,980. Both require strong math and problem-solving skills, but the differences are education, size, and focus (BLS, 2026). Research scientists usually need a master's degree to work in small but growing fields and focus on extensive research. In contrast, developers need a bachelor's degree, have far more openings each year, and build finished products for users. If I had to choose one, I am most definitely going toward software development because it offers a realistic path, and I can finish my courses at NVCC and then transfer to complete a bachelor's degree at GMU; then right after I can find an internship since the skills will be useful, and once I get myself financially established, I could then pursue a master's degree and move into research later, if time allows it.

References

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